Association Between Universal Face Shield in a Quaternary Care Center and Reduction of SARS-COV2 Infections Among Healthcare Personnel and Patients

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Disclosures

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- No relevant financial disclosures or conflict of interest

Introduction

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- SARS-COV2 transmission to healthcare personnel (HCP) and hospitalized patients is a significant challenge.
- Our hospital is a quaternary healthcare system with more than 500 beds and 8,000 HCP.
- Between April 1 and April 17, 2020, we implemented several infection prevention measures including:
  - Limiting entry to facility
  - Screening for symptoms and temperature
  - Universal masking for HCP
  - Universal masking for patients
  - Social distancing (avoid having lunch with other people)
  - Limited meeting size to less than 10

Surveillance testing

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- As a part of our pandemic response, we started a surveillance program for HCP and patients (started April 17). This program included:
  - Biweekly testing for HCP in high risk units (ED, transplant units, and COVID-19 units)
  - Weekly testing for HCP in cluster areas (>= 3 cases of HCP with COVID-19 diagnosis or any case of hospital-acquired infection [HAI])
  - Testing all asymptomatic patients on admission, prior to invasive procedures and every 7 days
  - Testing was voluntary for HCP and patients
  - HCP in other areas were allowed to be tested if desired or exposure history
  - HCP or patients with previous positive diagnosis were excluded

Measures

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- Percent of HCP testing positive for COVID-19

- HAI rates per 10,000 patient days
  - Possible HAI: days 5-13 from admission
  - Probable HAI: >= days 14 from admission

Texas Reopening

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Phase 1 (May 1)

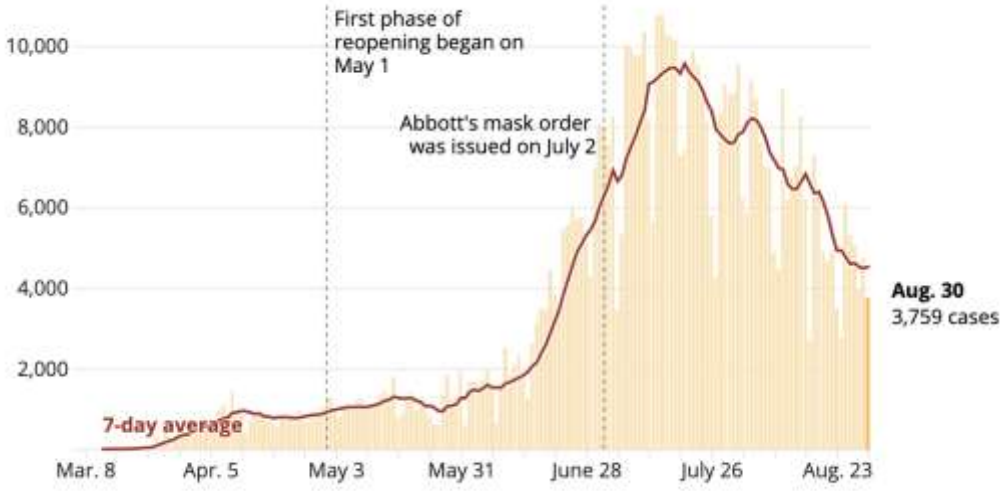
All retail stores, restaurants, movie theaters, museums, libraries, and malls (25% capacity)

Phase 2 (May 18-31)

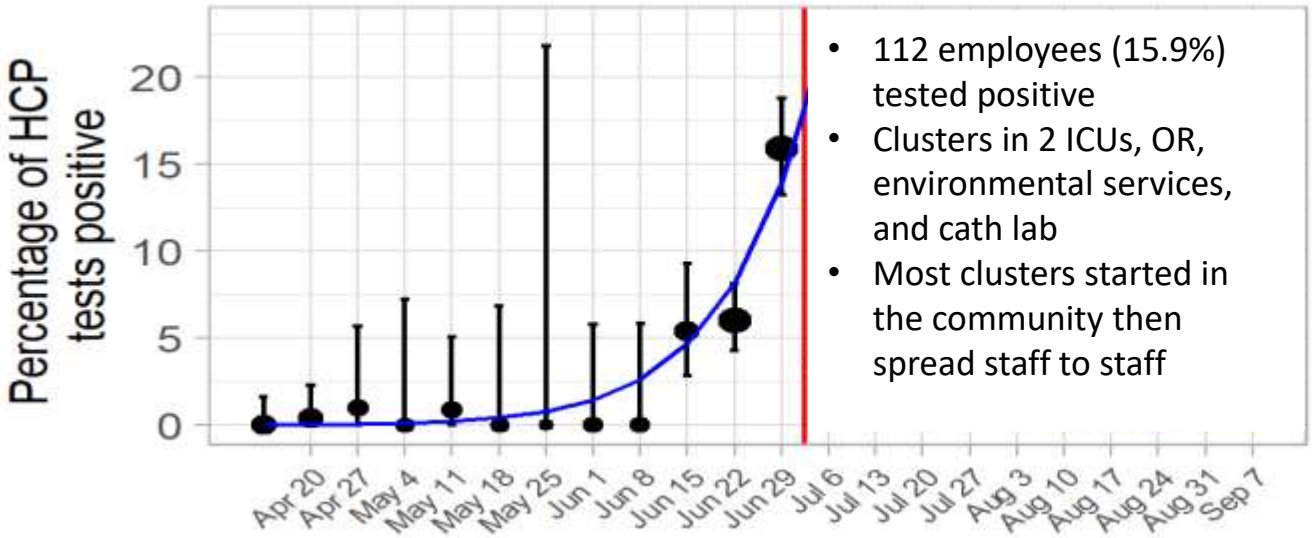
Child care centers, massage centers, youth clubs, bowling alleys, bingo Halls, simulcast racing, and skating rinks, bars, aquariums and natural caverns, zoos, and certain professional sports without in-person spectators. Restaurants capacity increased to 50%

Phase 3 (June 12)

Restaurants capacity increased to 75%, professional and college sports increased to 50% capacity



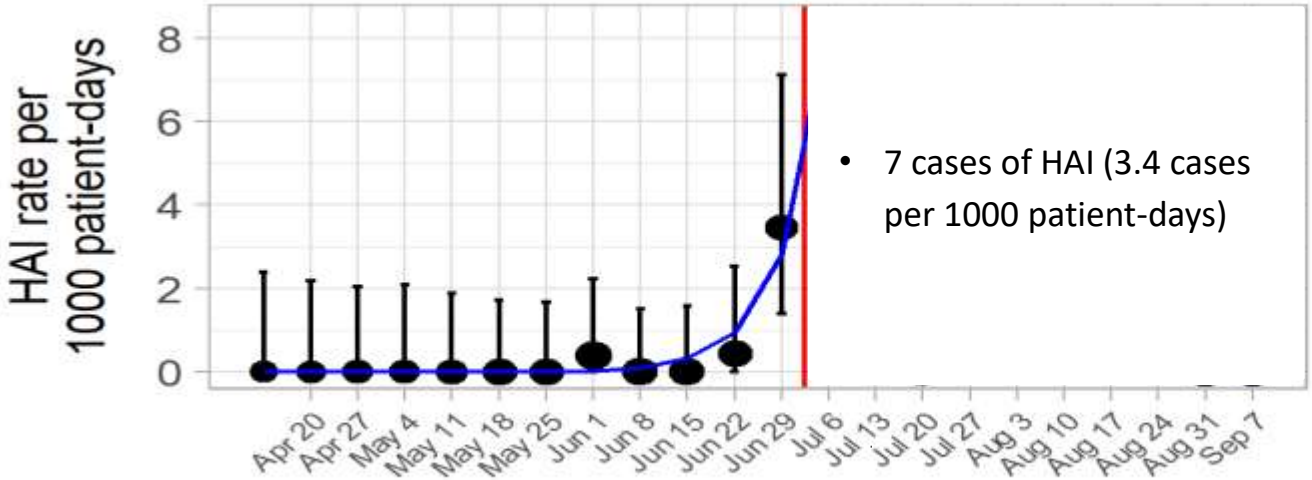
From New York Times



- 112 employees (15.9%) tested positive
- Clusters in 2 ICUs, OR, environmental services, and cath lab
- Most clusters started in the community then spread staff to staff

patients tested

- 250
- 500
- 750
- 1000



- 7 cases of HAI (3.4 cases per 1000 patient-days)

Patient-days

- 1000
- 1500
- 2000
- 2500

Texas Interventions

June 25- Texas closes bars, limits restaurant occupancy and cancels elective procedures in 4 counties

July 2 - Texas mandates face mask

July 9-Texas bans elective procedures in 100 counties

BSLMC Interventions

April 1- BSLMC implemented universal face masks for HCP

April 15- BSLMC implemented universal testing for HCP and patients

April 17- BSLMC implemented universal face masks for patients

July 6-BSLMC implemented universal face shields for HCP

Intervention 7/6

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- We implemented **universal face shield** for all healthcare personnel upon entry to facility.
- **Goggles were allowed as an alternative** if unable to tolerate face shields.

Methods

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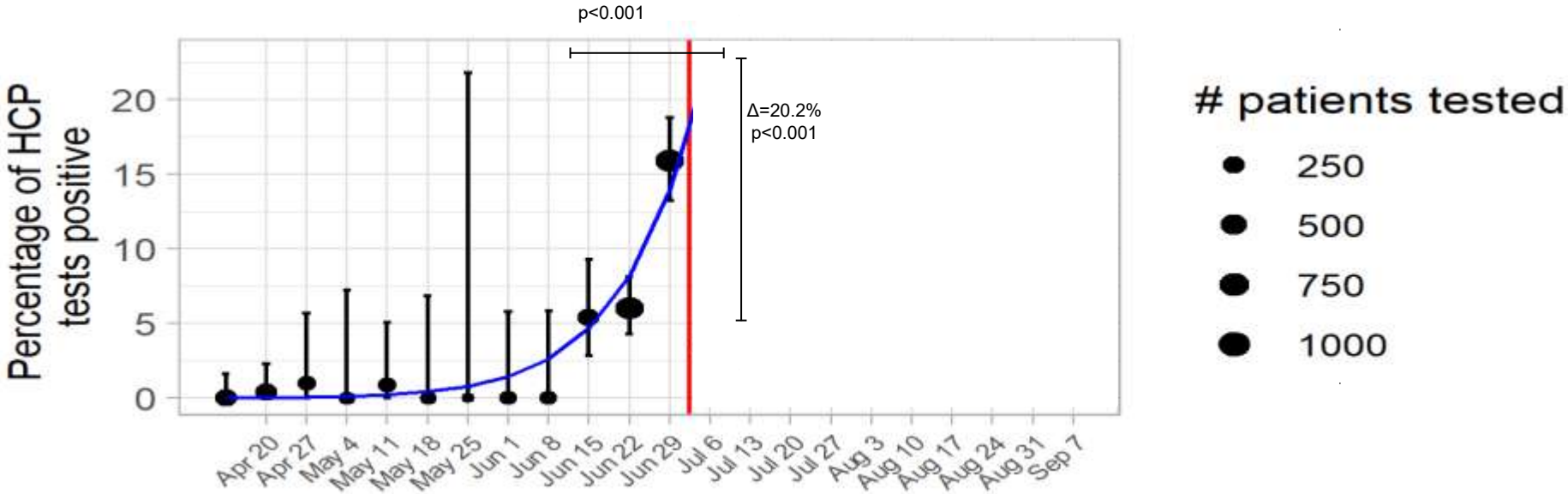
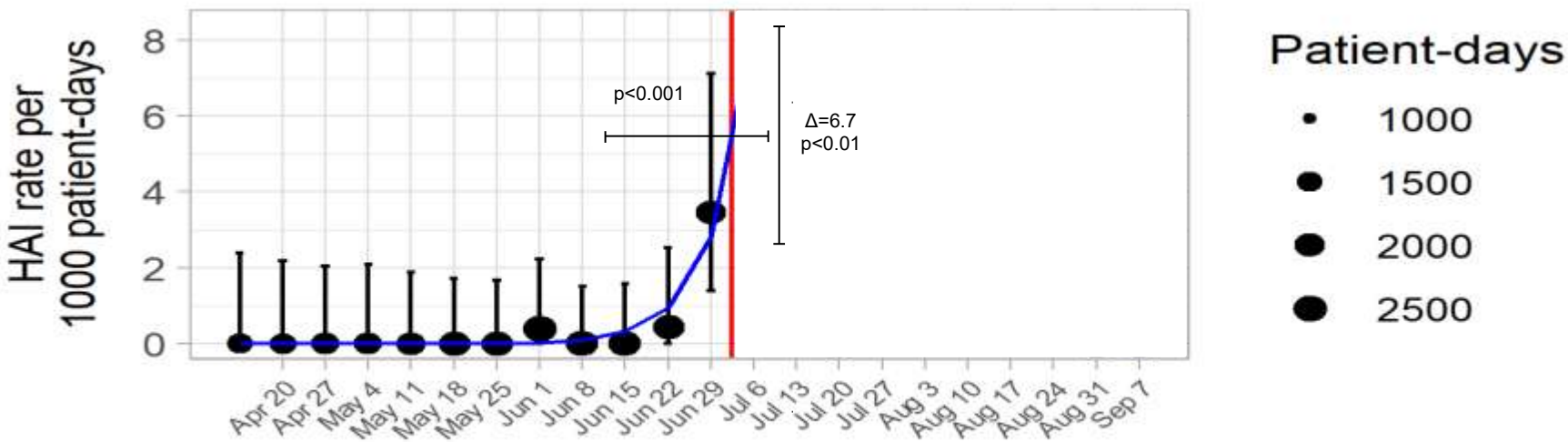
- We used R 4.0.2 and RStudio 1.3.1056 to perform the statistical analysis.
- We calculated the **proportion of positive HCW tests** with 95% CI and the **weekly rate of HAI per 1000 patient-days** with 95% CI using the *epitools* package (##version 0.5-10.1).
- We used **interrupted time series analysis with segmented regression** to examine the effect of our intervention on the difference in proportion of HCP positive for SARS-COV2 (using logistic regression) and possible or confirmed HAI (using Poisson regression).
- We defined significance as P values <0.05.
- This study was deemed exempt from IRB review.





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Results

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- Of 6527 HCP tested, 246 tested positive for SARS-COV2 (3.8%).
- In the preintervention period, the weekly positivity rate among HCP increased from 0% to 12.9%.
- During the intervention period, the weekly positivity rate among HCP decreased to 2.3%, with segmented regression showing a change in predicted proportion positive in week 13 (22.9% to 2.7%, $p<0.001$) and change in the post-intervention slope on the log odds scale ($p<0.001$).

Results

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- A total of 25 HAI cases were identified (16 possible and 9 confirmed).
- In the preintervention period, HAI cases increased from 0 to 7.
- During the intervention period, HAI cases decreased to 0.
- The change in predicted HAI rate due to the intervention was significantly different in week 13 with and without the intervention (8.4 vs 1.7 per 1000 patient-days, $p=0<0.01$), and change between pre-intervention and post-intervention slope on the log scale was also significant ($p<0.001$).

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Discussion

- Our study showed that the universal use of face shield was associated with:
  - A reduction in the percent of HCP testing positive for COVID-19
  - A reduction in HAI rates per 10,000 patient-days
- This could be explained by a reduction in viral transmission after universal face shield introduction.
- Similar results were described in community health workers in India after implementation of face shields[1]

1 Bhaskar ME, Arun S. SARS-CoV-2 infection among community health workers in India before and after use of face shields. JAMA. 2020 Aug 17.

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Limitations

- There were several state measures that were implemented during the time of the intervention that might have confounded the results.
- However, the number of SARS-COV2 continued to increase in the community during the first period of intervention despite these state measures.

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Conclusion

- Our results supports universal face shield use as a part of a multifaceted approach in areas of high SARS-COV2 community transmission.

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Acknowledgements

- Moses Saker-Valdez, RN
- Cindy Ly, MS, MLS(ASCP)^{CM}
- Ashlea Meier, MLS (ASCP)^{CM}
- Fengwei Zhong, MD
- Megan Fischer, MBA
- Takei Pipkins, MPH

Questions